

Name: Krishna RenjithDate: 5/29/2025Class: 4

MyDesign Portfolio

Element K Initial Template

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

- defining the problem (elements A-C)

It was easy to define the problem as the University of Maryland had provided us with information regarding our task, which was designing effective fire detection and suppression systems. After doing some research, it became clear that fire-related deaths and injuries are on the rise due to a lack of effective fire safety systems. One part which didn't go well when defining the problem was understanding what fire safety systems had been in use during these fire-related accidents. In most cases, the technology was either not present or not functioning. Therefore, we could not entirely gauge which system failures led to these incidents.

- ideation (element D)

One thing that went well was that every member of the group was able to come up with a solution which was planned on paper and then created on Planner 5D. This allowed us to create the prototype of the solution which everyone agreed would work best. One thing that didn't go well was that we did not know which materials would be provided by our instructor early on so we could not design the prototype accordingly.

- selection & prototype design (elements D-F)

One thing that went well was identifying which Common Core and NGSS standards backed our design requirements and design solutions. Another thing that went well was that once we acquired our design materials, it became easier to consider the design viability. One thing that didn't go well was developing corrective responses to our design review by the field expert. Our only field expert was Dr. Jim Milke, and as a result, we only

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came up with one corrective response.

- prototype building (element G)

One thing that went well was listing the materials we used and steps that we took in our project. Since each of the group members had a specific role (creating furniture, cutting cardboard, fastening screws, etc), it was easier to keep track of how much of a specific resource was used and how different parts of the prototype were built. Additionally, the main materials in our prototype were cardboard, glue, and popsicle sticks, which were easy to keep track of. One thing that did not go well was understanding the cost per unit of the materials we used. Since the materials were purchased by our teacher, we were not certain about the material costs.

- testing (elements H-I)

One thing that went well was determining how different parts of our house could be tested and whether those tests would result in a pass or fail. Most aspects of our house including the furniture, detection systems, and structure could be tested easily. However, it was difficult to test the suppression system because that would require setting the house on fire rather than holding a lighter against it. We should have included the suppression system (sprinklers) in our prototype so they could be tested beforehand.

- evaluation (element J)

One thing that went well was that we had come up with multiple stakeholders for our design in Element A including homeowners, firefighters, and children, who are all affected by the problem of rising fire related accidents due to a lack of effective fire detection and suppression systems. One thing which did not go well was that we only had one field expert and therefore, could not fully evaluate the limitations and positives of our design. Additionally, we did not meet with any of the stakeholders so we did not receive expert advice from them.

For each project step, explain: What went well? What didn't go well? Why?



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Lessons Learned About the Engineering Design Process (K3)

Based on my experience, designers need to understand what materials they are working with early on so that their design can be made accordingly. Multiple prototypes should be made so that further testing can occur even if the first prototype is destroyed. In our case, having multiple prototypes of the cardboard house would have allowed us to set one on fire in order to test the suppression systems which in our case, failed, due to a lack of testing. Additionally, extensive research on the problem should be done beforehand so that planning will be easier. Our team should have researched which fire systems are most likely to fail in houses which are affected by fire-related accidents. Also, A final suggestion would be to ensure that more than one field expert is present. Our group had only one field expert which made getting feedback difficult as he had to be present for the testing of all groups' designs. We did not have a field expert supervising our project but if future engineers have one, it will help them navigate the engineering design process with ease.

Element K: Reflection on the Design Project						
K1. Clear and comprehensive summary. The project designer provides _____ of the project.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	a robust and comprehensive summary of each step	a clear and comprehensive summary of the major steps	a generally clear and adequately-developed summary of most steps	a sparse summary of some steps	an overly-general summary	no summary
K2. Value judgment. A value judgment is _____ provided for project steps.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	consistently and robustly	consistently (for major steps)	consistently (for major steps, with one or two addressed in only a cursory manner)	partially (one or two addressed in only a cursory manner)	partially (and overly general or superficial)	not attempted

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			manner)			
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
K3. Inclusion of lessons learned. Lessons learned reflections are _____ 	substantive and are clearly useful to others attempting engineering design in the future	summarized and are clearly useful to others attempting engineering design in the future	summarized and at least most are useful to others attempting engineering design in the future	included in part and some are useful to others attempting engineering design in the future	minimally included and at least one is useful to others attempting engineering design in the future	not attempted, unclear and/or unlikely to be useful to others attempting engineering design in the future (or there is no evidence of lessons learned)